LATIHAN BOOLEAN ALGEBRA - CANONICAL & STANDARD FORMS

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PART I

3-1. Use truth table to prove the following.
a. $A'\cdot B' + A\cdot B' + A\cdot B = A + B'$
b. $A\cdot B\cdot C + A'\cdot B\cdot C + A'\cdot B\cdot C' = B\cdot C + A'\cdot B$

3-2. How many literals can be derived from n variables?
a. n b. $2n$ c. n^2 d. 2^n

3-3. How many minterms can be derived from 5 variables?
a. 5 b. 10 c. 25 d. 32

3-4. Which of the following is a sum term?
a. $A\cdot B$ b. $A'\cdot B$ c. $A+B'+C$ d. $A+B\cdot C$ e. $(A+B)\cdot C$

3-5. Which of the following is a sum-of-products expression?
a. $A\cdot B$ b. $A'\cdot B$ c. $A+B'+C$ d. $A+B\cdot C$ e. All above

3-6. Which of the following statements is valid?
a. $m43 + m34 = 0$
b. $A\cdot B + A'\cdot B' = 1$
c. $A'\cdot B'\cdot D + B'\cdot C\cdot D + A\cdot C\cdot D = A'\cdot B'\cdot D + A\cdot C\cdot D$
d. $A'\cdot B + B' = A' + B$
e. None of the above

3-7. Given a Boolean function $F(A,B,C) = \Sigma m(0, 5, 7)$. Which of the following is correct?
a. $F'(A,B,C) = \Sigma m(0, 5, 7)$
b. $F(A,B,C) = \Pi M(0, 5, 7)$
c. $F(A,B,C) = \Pi M(1, 2, 3, 4, 6)$
d. $F'(A,B,C) = \Sigma m(1, 2, 3)$
e. None of the above

3-8. Given a Boolean function $F(x,y,z) = y'\cdot(x+z') + x'\cdot z$. Which of the following is correct?
a. $F(x,y,z) = \Sigma m(0, 1)$
b. $F(x,y,z) = \Sigma m(0, 1, 2, 3, 4)$
c. $F(x,y,z) = \Sigma m(0, 1, 2, 4, 5)$
d. $F(x,y,z) = \Sigma m(0, 1, 3, 4, 5)$
e. $F(x,y,z) = \Sigma m(1, 2, 3, 4, 5)$

3-9. Given a Boolean function with 6 variables a, b, c, d, e, f. What is maxterm *M60*?

a. $a'\cdot b'\cdot c'\cdot d'\cdot e\cdot f$
b. $a'+b'+c'+d'$
c. $a'+b'+c'+d'+e+f$
d. $a\cdot b\cdot c\cdot d\cdot e'\cdot f'$
e. $a+b+c+d+e'+f'$

3-10. Given a Boolean function $F(a,b,c) = \Sigma m(0, 2, 5, 6, 7)$. If a=0, b=c=1, what is the value of F?

a. 0 b. 1 c. $a'\cdot b\cdot c$ d. a' e. unknown

3-11. Given a Boolean function $F(w,x,y,z) = w\cdot x\cdot z' + x\cdot y'\cdot z + x'\cdot z$. Which of the following minterms will appear in the sum-of-minterms expression of F?

a. $w\cdot x\cdot y\cdot z$ b. $w'\cdot x'\cdot y'\cdot z'$ c. $w'\cdot x'\cdot y\cdot z'$ d. $w\cdot x\cdot y'\cdot z$ e. $w\cdot x'\cdot y\cdot z'$

3-12. Given the following truth table of a Boolean function $F(x,y,z)$, which of the expressions represents F?

x	y	z	F
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	0

a. $x\cdot y'\cdot z' + x'\cdot y\cdot z + x\cdot z'$
b. $x'\cdot y\cdot z + x\cdot y' + x'\cdot y'\cdot z$
c. $x\cdot y'\cdot z + x'\cdot y'\cdot z + x'\cdot y$
d. $x\cdot y\cdot z + x'\cdot y'\cdot z + x\cdot y'$
e. None above

3-13. Given a Boolean function $F(A,B,Y,Z) = \Sigma m(0, 2, 5, 6, 8, 11, 15)$.

a. Express F in the canonical SOP form.
b. Express F in the canonical POS form.
c. Express F' in the little-m notation.
d. Express F' in the big-M notation.

3-14.* How many unique minterms does the following Boolean function contain?

$$\boldsymbol{F(A,B,C,D,E) = B\cdot D'\cdot E + A'\cdot C + A'\cdot B\cdot C\cdot D}$$

3-15.* What is $M30$ (maxterm 30) for a 5-variable Boolean function $F(A,B,C,D,E)$?

3-16. Using Boolean algebra, simplify the following expressions into the **SOP (sum-of-products) form**. You should attempt to make your answer as simplified as possible.

(a) $B\cdot D + B'\cdot(D + E) + D'\cdot(D + B)$
(b) $(B + B\cdot C)\cdot(B + B'\cdot C)\cdot(B + D)$
(c) $y\cdot z'\cdot(z' + z'\cdot x) + (x' + z')\cdot(x'\cdot y + x'\cdot z)$

3-17. State whether each of the following is a

i. sum-of-product (SOP) expression,
ii. canonical SOP expression,
iii. product-of-sum (POS) expression, or
iv. canonical POS expression.

If it is an SOP (or POS) expression, convert it to the canonical SOP (or canonical POS) expression, then express it in "little *m*" as well as "big *M*" notations, and draw the truth table using the derived expression. Is it possible to fill in the truth table without first obtaining the canonical forms? (Canonical SOP = sum-of-minterms; canonical POS = product-of-maxterms)

(a) $f(A,X,Z) = A \cdot X' \cdot Z + A' \cdot X' \cdot Z' + A \cdot X \cdot Z'$
(b) $f(p,q,r,s) = p \cdot q \cdot r' \cdot s + p' \cdot q' \cdot r \cdot s + p \cdot q' \cdot r \cdot s' + p' \cdot q \cdot r \cdot s' + p' \cdot q' \cdot r' \cdot s + p' \cdot q \cdot r' \cdot s'$
(c) $f(x,y,z) = x' + y + x' \cdot y \cdot z + x \cdot y \cdot z'$
(d) $f(A,B,C) = (A' + B + C) \cdot (A + B' + C) \cdot (A' + B + C')$
(e) $f(A,B,C) = (A + C) \cdot (B + C) \cdot (A' + B' + C)$

PART II

1. Manakah fungsi yang bukan **Canonical Forms**?

   a. F(a,b,c) = (a + b + c')(a' + b' + c)(a' + b + c')

   b. F(a,b,c) = (a + b)(c + d)

   c. F(x,y,z) = x + y' + z

   d. F(x,y,z) = xy'z + xyz' + x'y'z'

2. Suatu fungsi Boolean mengandung 5 variabel A, B, C, D, E. Berapakah index dari minterm A'.B.C'.D.E'?

   _____

3. Suatu fungsi Boolean mengandung 5 variabel A, B, C, D, E. Berapakah index dari maxterm (A + B' + C + D' + E') ?

   _____

4. Bentuk Product of Maxterms (POM) dari fungsi **F = ab' + b'c + ac** adalah:

   a. F = (a+b+c)(a+b+c')(a+b'+c)(a'+b'+c)

   b. F = (a+b+c')(a+b'+c')(a+b'+c)(a'+b'+c)

   c. F = (a+b+c)(a+b'+c')(a+b'+c)(a'+b'+c')

   d. F = (a+b+c)(a+b'+c)(a+b'+c')(a'+b'+c)

5. Bentuk Sum of Minterms (SOM) dari fungsi **F = (a+b')(a+c)(b'+c)** adalah:

   a. F = ab'c' + bc + a'b'c

   b. F = a'b'c' + a'b'c + ab'c + abc'

   c. F = abc + ab'c' + a'b'c + ab'c

   d. F = ab' + a'bc + abc

6. Lengkapilah Truth Table berikut dengan mengisi 0 atau 1 untuk **F = xy + yz + x'z.**

x	**y**	**z**	**F**
0	0	0	_____
0	0	1	_____
0	1	0	_____
0	1	1	_____
1	0	0	_____
1	0	1	_____
1	1	0	_____
1	1	1	_____

7. Tentukan bentuk Product of Maxterm dari: **F(x,y,z) = Σm(1,3,6,7)**

   Pilih salah satu:

   a. F = (x+y'+z)(x+y'+z)(x'+y+z)(x'+y+z')

   b. F = (x+y+z)(x'+y+z')(x+y'+z)(x'+y+z)

   c. F = (x+y+z)(x'+y+z')(x+y'+z')(x'+y+z)

   d. F = (x'+y+z)(x'+y'+z)(x+y'+z)(x'+y+z')

8. Tuliskan bentuk Product of Sum dari sum of minterms: **F = x'y'z'+xy'z'+xyz+xyz'**

   Pilih salah satu:

   a. F = (x + y')(y + z')

   b. F = (x + z)(x' + y')

   c. F = (x' + z)(x' + y)

   d. F = (x + y')(x' + z)

9. Tuliskan bentuk Sum of minterms dari **F(x,y,z) = ΠM(0,2,4,5)**

   Pilih salah satu:

   a. F = x'yz + x'y'z' + x'y'z + xyz'

   b. F = x'y'z + x'yz + xyz' + xyz

   c. F = x'yz + x'yz' + xyz + x'y'z

   d. F = x'yz + xyz' + xy'z + x'y'z

10. Manakah bentuk duality dari **F(X,Y,Z) = X'.(Y+Z').(X'+Z)**

    a. X.(Y'+Z).(X+Z')

    b. X' + Y.Z' + X'.Z

    c. X + Y'.Z + X.Z'

    d. X' + (Y.Z)' + (X'.Z)'

11. Tentukan bentuk paling sederhana dari fungsi boolean berikut hanya dalam 3 literal:

    F(X,Y,Z)=X'.Y'+X.Y.Z+X'.Y

    a. X.Y' + Z

    b. X'+Y.Z

    c. X'.Y+Z

    d. X'.Z + Y

12. Manakah complement dari **F (X,Y,Z) = (X+Z).X'.Y + X.Y.Z**

a. (X.Z + X' + Y) . (X + Y + Z)

b. (X'+Z').X.Y' + X'.Y'.Z'

c. (X'.Z' + X + Y') . (X' + Y' + Z')

d. ((X.Z)' + X' + Y) . (X + Y + Z)